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FEBRUARY 1944

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THE HISTORY
OF THE FUTURE
AND TODAY

LAST KNOWN
PAGES

EARTH

★ Femalhost

OFF THE BEAM

BY GEORGE D. SMITH

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*** START OF THE PROJECT GUTENBERG EBOOK OFF THE BEAM

Off the Beam

By GEORGE O. SMITH

Illustrated by Orban

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Thirty hours out of Mars for Terra, the *Solar Queen* sped along her silent, invisible course. No longer was she completely severed from all connection with the planets of the inner system; the trick cams that controlled the beams at Venus Equilateral kept the ship centered by sheer mathematics. It was a poor communications system, however, since it was but a one-way job. Any message-answering would have to be done thirty hours later when the ship made planetfall, and the regular terminal office of Interplanetary Communications could be employed.

In spite of her thirty hours at 2-G, which brought her velocity to eleven hundred miles per second, the beam-director cams did their job well enough. It was only in extreme cases of course-changing to dodge meteors that the beams lost the ship; since the cams were not clairvoyant, there was no way to know when the autopilot juggled the controls to miss a bit of cosmic dust. The cams continued to spear the space through which the ship was supposed to pass according to the course constants.

What made this trip ironic was the fact that Don Channing was aboard. The beams had been bombarding the *Solar Queen* continually ever since she left Mars with messages for the Director of Communications. In one sense, it seemed funny that Channing was for once on the end of a communications line where people could talk to him but upon which he could not talk back. On the other hand it was a blessing in disguise, for the Director of Communications was beginning to paper-talk himself into some means of contacting the Relay Station from a spaceship.

A steward found Channing in the salon and handed him a 'gram. Channing smiled, and the steward returned the smile and added: "You'll fix these ships to talk back one day. Wait until you read that one—you'll burn from here to Terra!"

"Reading my mail?" asked Channing cheerfully. The average spacegram was about as secret as a postcard, so Channing didn't mind. He turned the page over and read:

HOPE YOU'RE WELL FILLED WITH GRAVANOL AND
ADHESIVE TAPE FOR YOUR JUMP FROM TERRA TO

STATION. SHALL TAKE GREAT DELIGHT IN RIPPING
ADHESIVE TAPE OFF YOUR MEASLY BODY. LOVE.

ARDEN

"She will, too," grinned Don. "Well, I'd like to toss her one back, but she's got me there. I'll just fortify myself at the bar and think up a few choice ones for when we hit Mojave."

"Some day you'll be able to answer those," promised the steward. "Mind telling me why it's so tough?"

"Not at all," smiled Channing. "The problem is about the same as encountered by the old-time cowboy. It's a lot easier to hit a man on a moving horse from a nice, solid rock than it is to hit a man on a nice, solid rock from a moving horse. Venus Equilateral is quite solid as things go. But a spaceship's course is fierce. We're wobbling a few milliseconds here and a few there, and by the time you use that arc to swing a line of a hundred million miles, you're squirting quite a bit of sky. We're tinkering with it right now, but so far we have come up with nothing. Ah, well, since the human race got along without electric lights for a few million years, we can afford to tinker with an idea for a few months. Nobody is losing lives or sleep because we can't talk to the boys back home."

"We've been hopping from planet to planet for quite a number of years too," said the steward. "Quite a lot of them went by before it was even possible to contact a ship in space."

"And that was done because of an emergency. Probably this other thing will go on until we hit an emergency; then we shall prove that old statement about a loaf of bread being the maternal parent of a locomotive." Channing lit a cigarette, and puffed deeply. "Where do we stand?"

"Thirty hours out," answered the steward. "About ready for turnover. I imagine that the poor engineer's gang is changing cathodes about now."

"It's a long drag," said Channing. He addressed himself to his glass and began to think of a suitable answer for his wife's latest thrust.

Bill Hadley, of the power engineer's gang, spoke to the pilot's greenhouse below the ship. "Hadley to pilot room: Cathodes 1 and 3 ready."

"Pilot Greenland to Engineer Hadley: Power fade-over from even to odd now under way. Tubes 2 and 4 now dead; load on 1 and 3. You may enter 2 and 4."

"Check!"

Hadley cracked an air valve beside a circular air door. The hiss of entering air crescendoed and died, and then Hadley cracked the door that opened in upon the huge driver tube. With casual disregard for the annular electrodes that filled the tube and the sudden death that would come if the pilot sent the driving voltages surging into the electrodes, Hadley climbed to the top of the tube and used a spanner to remove four huge bolts. A handy differential pulley permitted him to lower the near-exhausted cathode from the girders to the air door where it was hauled to the deck. A fresh cathode was slung to the pulley and hoisted to place. Hadley bolted it tight and clambered back into the ship. He closed the air door and the valve, and then opened the valve that led from the tube to outer space. The tube evacuated and Hadley spoke once more to the pilot room.



"Hadley to Greenland: Tube 4 ready."

"Check."

The operation was repeated on Tube 2, and then Pilot Greenland said: "Fade-back beginning. Power diminishing on 1 and 3, increasing on 2 and 4. Power equalized, acceleration 2-G as before. Deviation from norm: two-tenths-G."

Hadley grinned at the crew. "You'd think that Greenland did all that himself, the way he talks. If it weren't for autopilots, we'd have been all over the sky."

Tom Bennington laughed. He was an old-timer, and he said in a reminiscent tone: "I remember when we used to do that on manual. There were as many cases of *mal de void* during cathode change as during turnover. Autopilots are the nuts—look! We're about to swing right now, and I'll bet a fiver that the folks below won't know a thing about it."

A coincidence of mammoth proportions occurred at precisely that instant. It was a probability that made the chance of drawing a royal flush look like the chances of tomorrow coming on time. It was, in fact, one of those things that they said couldn't possibly happen, which went to prove only how wrong they were. It hadn't happened yet and probably wouldn't happen again for a million million years, but it did happen once.

Turnover was about to start. A relay circuit that coupled the meteor-spotter to the autopilot froze for a bare instant, and the coincidence happened between the freezing of the relay contacts and the closing of another relay whose purpose it was to shunt the coupler circuits through another line in case of relay failure. In the inconceivable short time between the failure and the device that corrected failure, the *Solar Queen* hit a meteor head on.

It is of such coincidences that great tragedies and great victories are born.

The meteor, a small one as cosmic objects go, passed in through the broad observation dome at the top of the ship. Unhampered, it zipped through the central well of the *Solar Queen* and passed out through the pilot's greenhouse at the bottom of the ship. Its speed was nothing worth noting; a scant twenty miles per second almost sunward. But the eleven hundred miles per second of the *Solar Queen* made the passage of the meteor through the six hundred feet of the ship's length of less duration than the fastest camera shutter.

In those microseconds, the meteor did much damage.

It passed through the main pilotroom cable and scrambled those circuits which it did not break entirely. It tore the elevator system from its moorings. It entered as a small hole in the observation dome and left taking the entire pilot's greenhouse and all of the complex paraphernalia with it.

The lines to the driver tubes were scrambled, and the ship shuddered and drove forward at 10-G. An inertia switch tried to function, but the resetting solenoid had become shorted across the main battery and the weight could not drop.

Air doors clanged shut, closing the central well from the rest of the ship and effectively sealing the well from the crew.

The lights in the ship flickered and died. The cable's shorted lines grew hot and fire crept along its length and threatened the continuity. The heat opened fire-quenching vents and a cloud of CO₂, emerged together with some of the liquid gas itself. The gas quenched the fire and the cold liquid cooled the cable. Fuses blew in the shorted circuits—

And the *Solar Queen* continued to plunge on and on at 10-G; the maximum possible out of her driving system.

The only man who remained aware of himself aboard the *Solar Queen* was the man who was filled with gravanol and adhesive tape. No other person expected to be hammered down by high acceleration. Only Channing, who was planning to leave Terra in his own little scooter, was prepared to withstand high G. He, with his characteristic hate of doing anything slowly, was ready to make the Terra to Venus Equilateral passage at 5- or 6-G.

It might as well have caught him, too. With all of the rest unconscious, hurt, or dead, he was alone and firmly fastened to the floor of the salon under eighteen hundred pounds of his own, helpless weight.

And as the hours passed, the *Solar Queen* was driving farther and farther from the imaginary spot that was the focus of the communicator beams from Venus Equilateral.

The newly-replaced cathodes in the driving tubes were capable of driving the ship for about two hundred G-hours at 1-G, before exhaustion to the point of necessary replacement for safety purposes. The proportion is not linear, nor is it a square-law, but roughly it lies in the region just above linear, so that the *Solar Queen* drove on and on through space for ten hours at 10-G before the cathodes died for want of emitting surface. They died, not at once, but in irregular succession so that when the last erg of power was gone from the ship it was zooming on a straight line tangent from its point of collision but rolling in a wild gyration through the void.

And twenty-five hundred miles per second added to her initial velocity of eleven hundred miles per second added up to thirty-six hundred miles per second. She should have had about seventy-five million miles to go at 2-G, to reach Terra in thirty hours from the halfway point where she turned ends to go into deceleration. Instead, the *Solar Queen* after ten hours of misdirected 10-G acceleration was thirty million miles on her way, or about halfway to Terra. Three hours later, driving free, the *Solar Queen* was passing Terra, having missed the planet by a few million miles.

Back in space, at an imaginary junction between the beams from Venus Equilateral and the course registered for the *Solar Queen*, Arden Channing's latest message was indicating all sorts of mild punishment for her husband when she got him home.

By the time that the *Solar Queen* should have been dropping out of the sky at Mojave Spaceport, the ship would be one hundred and ninety million miles beyond Terra and flirting with the imaginary line that marked the orbit of Mars.

That would be in seventeen hours.

Weightless, Channing pursued a crazy course in the salon of the spinning ship. He ached all over from the pressure, but the gravanol had kept his head clear and the adhesive tape had kept his body intact. He squirmed around in the dimness and could see the inert figures of the rest of the people who had occupied the salon at the time of the mishap. He became sick. Violence was not a part of Channing's nature—at least he confined his violence to those against whom he required defense. But he knew that many of those people who pursued aimless orbits in the midair of the salon with him would never set foot on solidness again.

He wondered how many broken bones there were among those who had lived through the ordeal. He wondered if the medical staff of one doctor and two nurses could cope with it.

Then he wondered what difference it made if they were to go on and on, and from that thought came the one he should have thought of first: How

were they to stop going on and on? Channing had a rough idea of what had happened. He knew something about the conditions under which they had been traveling, how long, and in which direction. It staggered him, the figures he calculated in his mind. It behooved him to do something.

He bumped an inert figure, and grabbed. One hand took the back of the head and came away wet and sticky. Channing retched, and then threw the inert man from him. He coasted back against a wall, and caught a handrail. Hand-over-hand he went to the door and into the hall. Down the hall he went to the passengers' elevator shaft and with no thought of what his action would have been on any planet, Channing opened the door and drove down the shaft for several decks. He emerged and headed for the sick ward.

He found the doctor clinging to his operating table with his knees and applying a bandage to one of his nurses' heads.

"Hello, Doc," said Channing. "Help?"

"Grab Jen's feet and hold her down," snapped the doctor.

"Bad?" asked Don as he caught the flailing feet.

"Seven stitches, no fracture," said the doctor.

"How's the other one?"

"Unconscious, but unharmed. Both asleep in bed, thank God. So was I. Where were—? You're Channing and were all doped up with gravanol and adhesive. Thank yourself a god for that, too. I'm going to need both of my nurses and we'll all need you."

"Hope I can do some good," said Don.

"You'd better. Or any good I can do will be wasted. Better start right now. Here," the doctor produced a set of keys, "these will unlock anything in the ship but the purser's safe. You'll need 'em. Now get along and do something and leave the body-mending to me. Scram!"

"Can you make out all right?"

"As best I can. But you're needed to get us help. If you can't, no man in the Solar System can. You're in the position of a man who can not afford to help in succoring the wounded and dying. It'll be tough, but there it is. Get cutting. And for Heaven's sake, get us two things: Light and a floor. I

couldn't do more than slap on tape whilst floating in air. See you later, Channing, and good luck."

The nurse squirmed, groaned, and opened her eyes. "What happened?" she asked, blinking into the doctor's flashlight.

"Tell you later, Jen. Get Fern out of her coma in the ward and then we'll map out a plan. Channing, get out of here!"

Channing got after borrowing a spare flashlight from the doctor.

He found Hadley up in the instrument room with a half dozen of his men. They were a mass of minor and major cuts and injuries, and were working under a single incandescent lamp that had been wired to the battery direct by means of spare cable. The wire went snaking through the air in a foolish, crooked line, suspended on nothing. Hadley's gang were applying first aid to one another and cursing the lack of gravity.

"Help?" said Channing.

"Need it or offer it?" asked Hadley with a smile.

"Offer it. You'll need it."

"You can say that again—and then pitch in. You're Channing, of Communications, aren't you? We're going to have a mad scramble on the main circuits of this tub before we can unwind it. I don't think there's an instrument working in the whole ship."

"You can't unravel the whole works, can you?"

"Won't try. About all we can do is replace the lighting system and hang the dead cathodes in again. They'll be all right to take us out of this cockeyed skew-curve and probably will last long enough to keep a half-G floor under us for tinkering, for maybe forty or fifty hours. Assistant Pilot Darlange will have to learn how to run a ship by the seat of his pants—as far as I can guess there isn't even a splinter of glass left in the pilot room—so he'll have to correct this flight by feel and by using a haywire panel."

"Darlange is a school-pilot," grinned one of Hadley's men.

"I know, Jimmy, but I've seen him work on a bum autopilot, and he can handle haywire all right. It'll be tough without Greenland, but Greenland—" Hadley let the sentence fall; there was no need to mention the fact that Greenland was probably back there with the rest of the wreckage torn from the *Solar Queen*.

Jimmy nodded, and the action shook him from his position. He grabbed at a roll of tape that was floating near him and let it go with a laugh as he realized it was too light to do him any good.

"Too bad that this gyration is not enough to make a decent gravity at the ends, at least," snorted Hadley. He hooked Jimmy by an arm and hauled the man back to a place beside him. "Now look," he said, "I can't possibly guess how many people are still in working condition after this. Aside from our taped and doped friend here, the only ones I have are we who were snoozing in our beds when the crush came. I'll bet a cooky that the rest of the crowd are all nursing busted ribs, and worse. Lucky that full-G died slowly as the cathodes went out; otherwise we'd all have been tossed against the ceilings with bad effects.

"Jimmy, you're a committee of one to roam the crate and make a list of everyone who is still in the running and those who can be given minor repairs to make them fit for limited work. Doc has a pretty good supply of Stader splints; inform him that these are only to be used on men who can be useful with them. The rest will have to take to plaster casts and the old-fashioned kind of fracture-support.

"Pete, you get to the executive deck and tell Captain Johansson that we're on the job and about to make with repairs. As power engineer, I've control of the maintenance gang too, and we'll collect the whole, hale, and hearty of Michaels' crew on our merry way.

"Tom, take three of your men and begin to unravel the mess with an eye toward getting us lights.

"Tony, you can do this alone since we have no weight. You get the stale cathodes from the supply hold and hang 'em back in the tubes.

"Channing, until we get a stable place, you couldn't do a thing about trying to get help, so I suggest that you pitch in with Bennington, there, and help unscramble the wiring. You're a circuit man, and though power-line stuff is

not your forte, you'll find that running a lighting circuit is a lot easier than neutralizing a microwave transmitter. Once we get light, you can help us haywire a control panel. Right?"

"Right. And as far as contacting the folks back home goes, we couldn't do a darned thing until the time comes when we should be dropping in on Mojave. They won't be looking for anything from us until we're reported missing; then I imagine that Walt Franks will have everything from a spinthariscopes to a gold-foil electroscope set up. Right now I'm stumped, but we have seventeen hours before we can start hoping to be detected. Tom, where do we begin?"

Bennington smiled inwardly. To have Don Channing asking him for orders was like having Captain Johansson request the batteryman's permission to change course. "If you can find and remove the place where the shorted line is, and then splice the lighting circuits again, we'll have a big hunk of our work done. The rest of us will begin to take lines off of the pilot's circuits right here in the instrument room so that our jury-controls can be hooked in. You'll need a suit, I think, because I'll bet a hat that the shorted line is in the well."

For the next five hours, the instrument room became a beehive of activity. Men began coming in dribbles, and were put to work as they came. The weightlessness gave quite a bit of trouble; had the instrument panels been electrically hot, it would have been downright dangerous since it was impossible to do any kind of work without periodically coming against bare connections. Tools floated around the room in profusion, and finally Hadley appointed one man to do nothing but roam the place to retrieve "dropped" tools. The soldering operations were particularly vicious, since the instinctive act of flinging excess solder from the tip of an iron made droplets of hot solder go zipping around the room to splash against something, after which the splashes would continue to float.

Men who came in seeking to give aid were handed tools and told to do this or that, and the problem of explaining how to free a frozen relay to unskilled help was terrific.

Then at the end of five hours, Channing came floating in to the instrument room. He flipped off the helmet and said to Hadley: "Make with the main switch. I think I've got it."

Throughout the ship the lights blinked on.

With the coming of light, there came hope also. Men took a figurative hitch in their belts and went to work with renewed vigor. It seemed as though everything came to a head at about this time, too. Hadley informed Darlange that his jury-control was rigged and ready for action, and about the same time, the galley crew came in with slender-necked bottles of coffee and rolls.

"It was a job, making coffee," grinned the steward. "The darned stuff wanted to get out of the can and go roaming all over the place. There isn't a one of us that hasn't got a hot coffee scar on us somewhere. Now if he"—nodding at Darlange—"can get this thing straightened out, we'll have a real dinner."

"Hear that, Al? All that stands between us and dinner is you. Make with the ship-straightening. Then we'll all sit around and wait for Channing to think."

"Is the ship's communicator in working order?" asked Darlange.

"Sure. That went on with the lights."

Darlange called for everyone in the ship to hold himself down, and then he tied his belt to the frame in front of the haywired panel. He opened the power on drivers 1 and 2, and the ship's floor surged ever so little.

"How're you going to know?" asked Hadley.

"I've got one eye on the gyro-compass," said Darlange. "When it stops turning, we're going straight. Then all we have to do is to set our bottom end along the line of flight and pack on the decel. Might as well do it that way since every MPS we can lose is to our advantage."

He snapped switches that added power to Driver 3. Gradually the gyro-compass changed from a complex rotation-progression to a simpler pattern, and eventually the simple pattern died, leaving but one freedom of rotation. "I'm sort of stumped," grinned Darlange. "We're now hopping along, but

rotating on our long axis. How we stop axial rotation with drivers set parallel to that axis I'll never guess."

"Is there a lifeship in working order?" asked Hadley.

"Sure."

"Tom, turn it against the rotation and apply the drivers on that until we tell you to stop."

An hour later the ship had ceased to turn. Then Darlange jockeyed the big ship around so that the bottom was along the line of flight. Then he set the power for a half-G, and everyone relaxed.

Ten minutes later Captain Johansson came in.

"You've done a fine job," he told Hadley. "And now I declare an hour off for dinner. Dr. MacLain has got a working medical center with the aid of a few people who understand how such things work, and the percentage of broken bones, though terrific in number, is being taken care of. The passengers were pretty restive at first, but the coming of light seemed to work wonders. This first glimmer of power is another. About nine or ten who were able to do so were having severe cases of skysickness." He smiled ruefully. "I'm not too sure that I like no-weight myself."

"Have you been in the observation dome?" asked Don.

"Yes. It's pierced, you know."

"Did the meteor hit the telescope?"

"No, why?"

"Because I'm going to have to get a sight on Venus Equilateral before we can do anything. We'll have to beam them something, but I don't know what right now."

"Can we discuss that over a dinner?" asked the captain. "I'm starved, and I think that the rest of this gang is also."

"You're a man after my own heart," laughed Channing. "The bunch out at the Station wouldn't believe me if I claimed to have done anything without drawing it up on a tablecloth."

"Now," said Channing over his coffee. "What have we in the way of electronic equipment?"

"One X-ray machine, a standard set of communicating equipment, one beam receiver with 'type machine for collecting stuff from your Station, and so on."

"You wouldn't have a betatron in the place somewhere?" asked Don hopefully.

"Nope. Could we make one?"

"Sure. Have you got about ten pounds of No. 18 wire?"

"No."

"Then we can't."

"Couldn't you use a driver? Isn't that some kind of beam?"

"Some kind," admitted Channing. "But it emits something that we've never been able to detect except in an atmosphere where it ionizes the air into a dull red glow."

"You should have been wrecked on the *Sorcerer's Apprentice*," laughed Hadley. "They're the guys who have all that kind of stuff."

"Have they?" asked Johannson.

"The last time I heard, they were using a large hunk of their upper hull for a VanDerGraf generator."

"That would do it," said Channing thoughtfully. "But I don't think I'd know how to modulate a VanDerGraf. A betatron would be the thing. You can modulate that, sort of, by keying the input. She'd give out with hundred-and-fifty-cycle stuff, but so what? We made the *Empress of Kolain* sit up and say uncle on hundred-cycle stuff. How much of a trick is it to clear the observation dome from the top?"

"What do you intend to do?"

"Well, we've got a long, hollow tube in this ship. Knock out the faceted dome above, and we can rig us up a huge electron gun. We'll turn the ship to

point at the Station and beam 'em a bouquet of electrons."

"How're you going to do that?"

"Not too tough, I don't think. Down here," and Channing began to trace on the tablecloth, "we'll put us a hot cathode. About this level we'll hang the first anode, and at this level we'll put the second anode. Here'll be an acceleration electrode, and up near the top we'll put a series of focusing anodes. We'll tap in to the driver-tube supply and take off voltages to suit us. Might use a tube at that, but the conversion to make an honest electron gun out of it would disrupt our power, and then it would be impossible to make a driver out of it again without recourse to a machine shop."

"How are you going to make electrodes?"

"We'll use the annular gratings that run around the central well at each level," said Channing. "We'll have a crew of men cut 'em free and insulate the resulting rings with something. Got anything?"

"There is a shipment of methyl-methacrylate rods for the Venus Power Co. in Hold 17," said the cargo master.

"Fine," said Channing. "What size?"

"Three inches by six feet."

"It'll be tricky work, and you'll have to wait until your cut edge has cooled before you hook on the rods," mused Don. "But that's the ticket."

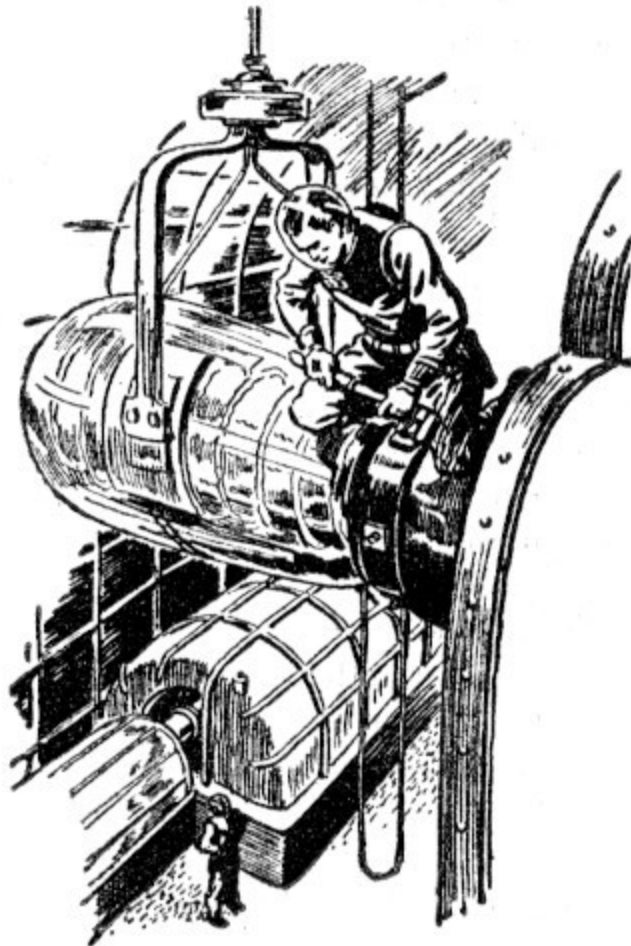
"Which floors do you want?"

"Have you got a scale drawing of the *Solar Queen*?"

"Sure."

"Then this is where my tablecloth artistry falls flat. The focusing of an electron beam depends upon the electrode spacing and the voltage. Since our voltage is fixed if we take it from the driver electrodes, we'll have to do some mighty fine figuring. I'll need that scale drawing."

Channing's tablecloth engineering was not completely wasted. By the time the scale drawing was placed before him, Channing had half of the table filled with equations. He studied the drawing, and selected the levels which were to serve as electrodes. He handed the drawing to Hadley, and the power engineer began to issue instructions to his gang.



Then the central well began to swarm with spacesuited men who bore cutting torches. Hot sparks danced from the cut girders that held the floorings, and at the same time, a crew of men were running cables from the

various levels to the instrument room. More hours passed while the circular sections were insulated with the plastic rods.

The big dome above was cut in sections and removed, and then the sky could be seen all the way from the bottom of the ship where the pilot's greenhouse should have been.

Channing looked it over and then remarked: "All we need now is an electron collector."

"I thought you wanted to shoot 'em off," objected Hadley.

"I do. But we've got to have a source of supply. You can't toss baseballs off of the Transplanet Building in Northern Landing all afternoon, you know, without having a few brought to you now and then. Where do you think they come from?"

"Hadn't thought of it in that way. What'd happen?"

"We'd get along for the first umpty-gillion electrons, and then all the soup we could pack on would be equalized by the positive charge on the ship and we couldn't shoot out any more until we got bombarded by the sun—and that bombardment is nothing to write home about as goes quantity. What we need is a selective solar intake plate of goodly proportions."

"We could use a mental telepathy expert, too. Or one of those new beams that Baler and Carroll dug up out of the Martian desert. I've heard that those things will actually suck power out of any source, and bend beams so as to enter the intake vent, or end."

"We haven't one of those, either. Fact of the matter is," grinned Channing ruefully, "we haven't much of anything but our wits."

"Unarmed, practically," laughed Hadley.

"Half armed, at least. Ah, for something to soak up electrons. I'm now wondering if this electron gun is such a good idea."

"Might squirt some protons out the other direction," offered Hadley.

"That would leave us without either," said Don. "We'd be like the man who tossed baseballs off of one side and himself off the other—Hey! Of course we have some to spare. We can cram electrons out of the business end, thus stripping the planetary rings from the atoms in our cathode. From the far

side we'll shoot the canal rays, which in effect will be squirting protons, or the nuclei. Since the planetaries have left for the front, it shouldn't be hard to take the protons away, leaving nothing. At our present voltages, we might be able to do it." Channing began to figure again, and he came up with another set of anodes to be placed beyond the cathode. "We'll ventilate the cathode and hang these negative electrodes on the far side. They will attract the protons, impelled also by the positive charge on the front end. We'll maintain a balance that way, effectively throwing away the whole atomic structure of the cathode. The latter will fade, just as the cathodes do in the driving tubes, only we'll be using electronic power instead of sub-electronic. Y'know, Hadley, some day someone is going to find a way to detect the—we'll call it radiation for want of anything better—of the driver. And then there will open an entirely new field of energy. I don't think that anybody has done more about the so-called sub-electronic field than to make a nice, efficient driving device out of it.

"Well, let's get our canal-ray electrodes in place. We've got about two hours before they realize that we aren't going to come in at Mojave. Then another two hours worth of wild messages between the Relay Station and Mojave. Then we can expect someone to be on the lookout. I hope to be there when they begin to look for us. At our present velocity, we'll be flirting with the Asteroid Belt in less than nothing flat. That isn't too bad—normally—but we're running without any meteor detector and autopilot coupler. We couldn't duck anything from a robin's egg on up."

"We'll get your anodes set," said Hadley.

Walt Franks grinned at Arden Channing. "That'll burn him," he assured her.

"It's been on the way for about twenty minutes," laughed Arden. "I timed it to arrive at Terra at the same time the *Solar Queen* does. They'll send out a special messenger with it, just as Don is getting aboard his little scooter. It'll be the last word, for we're not following him from Terra to here."

"You know what you've started?" asked Franks.

"Nothing more than a little feud between husband and self."

"That's just the start. Before he gets done, Don will have every ship capable of answering back. I've found that you can catch him off base just once. He's a genius—one of those men who never make the same mistake twice. He'll never again be in a position to be on the listening end only."

"Don's answer should be on the way back by now," said Arden. "Could be you're right. Something should be done."

"Sure I'm right. Look at all the time that's wasted in waiting for a landing to answer 'grams. In this day and age, time is money, squared. The latter is to differentiate between this time and the first glimmering of speedy living."

"Was there a first glimmering?" asked Arden sagely. "I've often thought that the speed-up was a stable acceleration from the dawn of time to the present."

"All right, go technical on me," laughed Walt. "Things do move. That is, all except that message from your loving husband."

"You don't suppose he's squelched?"

"I doubt it. Squelching Donald Channing is a job for a superbeing. And I'm not too sure that a superbeing could squelch Don and make him stay squelched. Better check on Mojave."

"Gosh, if Don missed the *Solar Queen* and I've been shooting him all kinds of screwy 'types every hour on the hour; Walt, that'll keep him quiet for a long, long time."

"He'd have let you know."

"That wouldn't have been so bad. But if the big bum missed and was ashamed of it—that'll be the pay-off. Whoa, there goes the 'type!"

Arden drew the tape from the machine:

MESSAGE BEING HELD FOR ARRIVAL OF *SOLAR QUEEN*.

Walt looked at his watch and checked the course constants of the *Solar Queen*. He called the beam-control dome and asked for the man on the *Solar Queen* beam.

"Benny," he said, "has the *Solar Queen* arrived yet?"

"Sure," answered Benny. "According to the mechanical mind here, they've been on Mojave for twenty minutes."

"Thanks." To Arden he said: "Something's strictly fishy."

Arden sat at the machine and pounded the keys:

SOLAR QUEEN DUE TO ARRIVE AT 19:06:41. IT IS NOW
19:27:00. BEAM CONTROL SAYS TRANSMISSIONS
ENDED BECAUSE OF COINCIDENCE BETWEEN TERRA
BEAM AND STATION-TO-SHIP BEAM. PLEASE CHECK.

Arden fretted and Walt stamped up and down the room during the long minutes necessary for the message to reach Terra and the answer to return. It came right on the tick of the clock:

HAVE CHECKED COURSE CONSTANTS. SOLAR QUEEN
OVER-DUE NOW FIFTY MINUTES. OBVIOUSLY
SOMETHING WRONG. CAN YOU HELP?

Walt smiled in a grim fashion. "Help!" he said. "We go on and on for years and years with no trouble—and now we've lost the third ship in a row."

"They claim that those things always run in threes," said Arden. "What are we going to do?"

"I don't know. We'll have to do something. Funny, but the one reason we must do something is the same reason why something can be done."

"I don't get that."

"With Channing on the *Solar Queen*, something can be done. I don't know what, but I'll bet you a new hat that Don will make it possible for us to detect the ship. There is not a doubt in my mind that if the ship is still spaceworthy, we can narrow the possibilities down to a thin cone of space."

"How?"

"Well," said Franks, taking the fountain pen out of the holder on the desk and beginning to sketch on the blotter, "the course of the *Solar Queen* is not a very crooked one, as courses go. It's a very shallow skew curve. Admitting the worst, collision, we can assume only one thing. If the meteor were small enough to leave the ship in a floating but undirigible condition,

it would also be small enough to do nothing to the general direction of the ship. Anything else would make it useless to hunt, follow?"

"Yes, go on."

"Therefore we may assume that the present position of the *Solar Queen* is within the volume of a cone made by the tangents of the outermost elements of the space curve that is the *Solar Queen's* course. We can take an eight-thousand-mile cylinder out of one place—for the origin of their trouble is between Mars and Terra and the 'shadow' of Terra in the cone will not contain the *Solar Queen*."

"Might have passed close enough to Terra to throw her right into the 'shadow' of Terra by attraction," objected Arden.

"Yeah, you're right. O.K., so we can't take out that cylinder of space. And we add a sort of side-wise cone on to our original cone, a volume through which the *Queen* might have flown after passing close enough to Terra to be deflected. I'll have the slipstick experts give a guess as to the probability of the *Queen's* course, and at the same time we'll suspend all incoming operations. I'm going to set up every kind of detector I can think of, and I don't want anything upsetting them."

"What kind of stuff do you expect?" asked Arden.

"I dunno. They might have a betatron aboard. In that case we'll eventually get a blast of electrons that'll knock our front teeth out. Don may succeed in tinkering up some sort of electrostatic field. We can check the solar electrostatic field to about seven decimal places right here, and any deviation in the field to the tune of a couple of million electron volts at a distance of a hundred million miles will cause a distortion in the field that we can measure. We'll ply oscillating beams through the area of expectation and hope for an answering reflection, though I do not hope for that. We'll have men on the lookout for everything from smoke signals to helio. Don't worry too much, Arden, your husband is capable of doing something big enough to be heard. He's just the guy to do it."

"I know," said Arden soberly. "But I can't help worrying."

"Me, too. Well, I'm off to set up detectors. We'll collect something."

"Have we got anything like a piece of gold leaf?" asked Channing.

"I think so, why?"

"I want to make an electroscope. That's about the only way I'll know whether we are getting out with this cockeyed electron gun."

"How so?" asked Hadley.

"We can tell from the meter that reads the beam current whether anything is going up the pipe," explained Channing. "But if we just build us up a nice heavy duty charge—as shown by the electroscope—we'll be sure that the electrons are not going far. This is one case where no sign is good news."

"I'll have one of the boys set up an electroscope in the instrument room."

"Good. And now have the bird on the telescope forget trying to find Venus Equilateral by dead reckoning and sight. Have him set the scope angles to the figures here, and then have him contact Darlange and have the ship slued around so that Venus is on the cross hairs. That'll put us on a line for the Station by a few thousand miles. We can afford to miss. A bundle of electrons of our magnitude zipping past the detectors that Walt can set up will make a reading."

Hadley called the observation dome. "Tim," he said, giving a string of figures, "set your 'scope for these and then get Darlange to slue the crate around so that your cross hairs are on Venus."

"O.K.," answered Tim. "That's going to be a job. This business of looking through a 'scope while dressed in a spacesuit is no fun. Here goes."

He called Darlange, and the communicator system permitted the men in the instrument room to hear his voice. "Dar," he said, "loop us around about forty-one degrees from Driver 3."

Darlange said: "Right!" and busied himself at his buttons.

"Three degrees on Driver 4."

"Right."

"Too far, back her up a degree on 4."

Darlange laughed. "What do you think these things are, blocks and tackles? You mean: 'Compensate a degree on 2.'"

"You're the pilot. That's the ticket—and I don't care if you lift it on one hand. Can you nudge her just a red hair on 3?"

"Best I can do is a hair and a half," said Darlange. He gave Driver 3 just a tiny, instantaneous surge.

"Then take it up two and back one and a half," laughed Tim. "Whoa, Nellie, you're on the beam."

"Fine."

"O.K., Dar, but you'll have to play monkey on a stick. I'll prime you for any moving so that you can correct immediately."

"Right. Don, we're on the constants you gave us. What now?"

"At this point I think a short prayer would be of assistance," said Channing soberly. "We're shooting our whole wad right now."

"I hope we make our point."

"Well, it's all or nothing," agreed Don as he grasped the switch.

He closed the switch, and the power demand meters jumped up across their scales. The gold-leaf electroscope jumped once; the ultra-thin leaves jerked apart by an inch, and then oscillated stiffly until they came to a balance. Channing, who had been looking at them, breathed deeply and smiled.

"We're getting out," he said.

"Can you key this?" asked Hadley.

"No need," said Channing. "They know we're in the grease. We know that if they can collect us, they'll be on their way. I'm going to send out for a half-hour, and then resort to a five-minute transmission every fifteen minutes. They'll get a ship after us with just about everything we're likely to need, and they can use the five-minute transmissions for direction finding. The initial shot will serve to give them an idea as to our direction. All we can do now is to wait."

"And hope," added Captain Johansson.

Electrically, Venus Equilateral was more silent than it had ever been. Not an electrical appliance was running on the whole station. People were cautioned about walking on deep-pile rugs, or combing their hair with plastic combs, or doing anything that would set up any kind of electronic charge. Only the highly filtered generators in the power rooms were running and these had been shielded and filtered long years ago; nothing would emerge from them to interrupt the ether. All incoming signals were stopped.

And the men who listened with straining ears claimed that the sky was absolutely clear save for a faint crackle of cosmic static which they knew came from the corona of the sun.

One group of men sat about a static-field indicator and cursed the minute wiggling of the meter, caused by the ever-moving celestial bodies and their electronic charges. A sunspot emission passed through the Station once, and though it was but a brief passage, it sent the electrostatic field crazy and made the men jump.

The men who were straining their ears to hear became nervous, and were jumping at every loud crackle.

And though the man at the telescope knew that his probability of picking up a sight of the *Solar Queen* was as slender as a spider's web, he continued to search the starry heavens. He swept the narrow cone of the heavens wherein the *Solar Queen* was lost according to the mathematical experts, and he looked at every bit of brightness in the field of his telescope as though it might be the missing ship.

The beam-scanners watched their return-plates closely. It was difficult because the receiver gains were set to maximum, and every tick of static caused brief flashes of light upon their plates. They would jump at such a flash and watch for it to reappear on the next wipe, for a continuous spot of light indicated the ship they sought. Then, as the spot did not reappear, they would go on with their beams to cover another infinitesimal portion of the sky. Moving forward across the cone of expectancy bit by bit, they crossed and recrossed until they were growing restive.

Surely the ship must be there!

At the South End landing stage, a group of men were busy stocking a ship. Supplies and necessities were carried aboard, while another group of men

tinkered with the electrical equipment. They cleared a big space in the observation dome, and began to install a replica of the equipment used on the Station for detection. No matter what kind of output Channing sent back, they would be able to follow it to the bitter end.

They made their installations in duplicate, with one piece of each equipment on opposite sides of the blunt dome. Balancing the inputs of each kind by turning the entire ship would give them a good indication of direction.

Franks did not hope that the entire installation could be completed before the signal came, but he was trying to outguess himself by putting some of everything aboard. When and if it came, he would be either completely ready with everything or he at least would have a good start on any one of the number of detectors. If need be, the equipment from the Station itself could be removed and used to complete the mobile installation.

Everything was in a complete state of nervous expectancy. Watchers watched, meter readers squinted for the barest wiggle, audio observers listened, trying to filter any kind of man-made note out of the irregular crackle that came in.

And the Station announcing equipment was dead quiet, to be used only in case of emergency or to announce the first glimmer of radiation, whether it be material, electrical, kinetic, potential, or wave front.

Long they listened—and then it came.

The Station announcing equipment broke forth in a multitude of voices.

"Sound input on radio."

"Visual indication on scanner plates!"

"Distortion on electrostatic field indicator."

"Super-electroscopes indicate negative charge!"

"Nothing on the telescope!"

There were mingled cheers and laughter as the speaker system broke away from its babel, and each group spoke its piece with no interference. Walt Franks left the ship at the South End and raced to the Beam Control dome, just as fast as the runway car would take him. He ran into the dome in spacesuit and flipped the helmet back over his shoulders. "What kind of indication?" he yelled.

Men crowded around him, offering him papers and shouting figures.

"Gosh," he said, "Don can't have everything going up there."

"He's hit just about everything but the guy squinting through the 'scope."

"What's he doing?" asked Franks of no one in particular.

One of the radiation engineers who had been busy with the electrostatic field indicator said: "I think maybe he's using some sort of electron gun—like the one you tried first off on the meteor-destroyer-job, remember?"

"Yeah, but that one wouldn't work—unless Don has succeeded in doing something that we couldn't do. Look, Charley, we haven't had time to set up a complete field indicator on the ship. Grab yours and give the boys a lift installing it, hey?"

"Sure thing."

"And look, fellows, any indication of direction, velocity, or distance?"

"Look for yourself," said the man on the beam scanner. "The whole plate is shining. We can't get a fix on them this way—they're radiating themselves and that means that our scanner-system finder is worthless."

"We can, but it's rough," offered one of the radio men. "It came from an area out beyond Terra—and as for our readings it might have covered a quarter of the sky."

"The field indicator is a short-base finder," explained Charley. "And no less rough than the radio boys. I'd say it was out beyond Terra by fifty million miles at least."

"Close enough. We'll have to track 'em down like a radio-equipped bloodhound. Charley, come along and run that mechanico-electro-monstrosity of yours. Gene, you can come along and run the radio finder. Oh yes, you, Jimmy, may continue to squint through that eyepiece of yours

—but on the *Relay Girl*. We need a good, first-class squinter, and you should have an opportunity to help."

Jimmy laughed shortly. "The only guy on the Station that didn't get an indication was me. Not even a glimmer."

"Channing didn't know we'd be *looking* for him, or he'd probably light a flare, too. Cheer up, Jimmy, after all this crude, electrical rigamarole is finished, and we gotta get right down to the last millimeter, it's the guy with the eye that polishes up the job. You'll have your turn."

Twenty minutes after the first glimmer of intelligent signal, the *Relay Girl* lifted from the South End and darted off at an angle, setting her nose roughly in the direction of the signal.

Her holds were filled with spare batteries and a whole dozen replacement cathodes as well as her own replacements. Her crew was filled to the eyebrows with gravanol, and there must have been a mile of adhesive tape and cotton on their abdomens. At 6-G she left, and at 6-G she ran, her crew immobilized but awake because of the gravanol. And though the acceleration was terrific, the tape kept the body from folding of its own weight. When they returned, they would all be in the hospital for a week, but their friends would be with them.

Ten minutes after take-off, the signals ceased.

Walt said: "Keep her running. Don's saving electricity. Tell me when we pick him up again."

Franklen, the pilot, nodded. "We haven't got a good start yet. It'll be touch and go. According to the slipstick boys, they must be clapping it up at between twenty-five hundred and five thousand miles per second to get that far—and coasting free or nearly so. Otherwise they'd have come in. Any suggestions as to course?"

"Sure. Whoop it up at six until we hit about six thousand. Then decelerate to four thousand by using 1-G. We'll vacillate in velocity between four and five until we get close."

Forty-one hours later, the *Relay Queen* made turnover and began to decelerate.

Channing said to Captain Johannson: "Better cut the decel to about a quarter-G. That'll be enough to keep us from bumping our heads on the ceiling and it will last longer. This is going to be a long chase, and cutting down a few MPS at a half-G isn't going to make much never-mind. I'll hazard a guess that the boys are on their way right now."



"If you say so," said Johannson. "You're the boss from now on. You know that wild bunch on the Station better than I do. For myself, I've always felt that an answer was desirable before we do anything."

"I know Franks and my wife pretty well—about as well as they know me. I've put myself in Walt's place—and I know that Walt would do. So—if Walt didn't think of it, Arden would—I can assume that they are aware of us, have received our signals, and are, therefore, coming along as fast as they can. They'll come zipping out here at from five- to seven-G to what they think is halfway and then decelerate again to a sane velocity. We won't catch sight of them for sixty or seventy hours, and when we do, they'll be going so fast that it will take another twenty hours worth of manipulation to match their speed with ours. Meanwhile, I've got the gun timed to shoot our signal. When the going gets critical, I'll cut the power and make it continuous."

"You're pretty sure of your timing?"

"Well, the best they can do as for direction and velocity and distance is a crude guess. They'll place us out here beyond Terra somewhere. They'll calculate the course requirements to get us this far in the time allotted, and come to a crude figure. I'd like to try keying this thing, but I know that keying it won't work worth a hoot at this distance. Each bundle of keyed electrons would act as a separate negative charge that would spread out and close up at this distance. It's tough enough to hope that the electron beam will hold together that far, let alone trying to key intelligence with it. We'll leave well enough alone—and especially if they're trying to get a fix on us; there's nothing worse than trying to fix an intermittent station. Where are we now?"

"We're on the inner fringe of the Asteroid Belt, about thirty million miles North, and heading on a secant course at thirty-four hundred MPS."

"Too bad Jupiter isn't in the neighborhood," said Channing. "We'll be flirting with his orbit by the time they catch us."

"Easily," said Johannson. "In sixty hours, we'll have covered about six hundred and fifty million miles. We'll be nearer the orbit of Saturn, in spite of the secant course."

"Your secant approaches a radius as you get farther out," said Don, absently. "As far as distances go. Ah, well, Titan, here we come!"

Johannson spoke to the doctor. "How're we doing?"

"Pretty well," said Doc. "There's as pretty an assortment of fractured ribs, broken limbs, cracked clavicles, and scars, mars, and abrasions as you ever saw. There are a number dead, worse luck, but we can't do a thing about them. We can hold on for a week as far as food and water goes. Everyone is now interested in the manner of our rescue rather than worrying about it." He turned to Channing. "The words Channing and Venus Equilateral have wonderful healing powers," he said. "They all think your gang are part magician and part sorcerer."

"Why, for goodness' sake?"

"I didn't ask. Once I told 'em you had a scheme to contact the Relay Station, they were all satisfied that things would happen for the better."

"Anything we can do to help you out?"

"I think not," answered Doc. "What I said before still goes. Your job is to bring aid—and that's the sum total of your job. Every effort must be expended on that and that alone. You've got too many whole people depending on you to spend one second on the hurt. That's my job."

"O.K.," said Channing, "But it's going to be a long wait."

"We can afford it."

"I hope we're not complicating the job of finding us by this quartering deceleration," said Johannson.

"We're not. We're making a sort of vector from our course, but the deviation is very small. As long as the fellows follow our radiation, we'll be found," Channing said with a smile. "The thing that is tough is the fact that all the floors seem to lean over."

"Not much, though."

"They wouldn't lean at all if we were running with the whole set of equipment," said Darlange. "We run a complete turnover without spilling a drop from the swimming pool."

"Or even making the passengers aware of it unless they're looking at the sky."

"Stop worrying about it," said Doc. "I'm the only guy who has to worry about it and as long as the floor is still a floor, I can stand sliding into the corner once in a while."

"We might tinker with the turnover drivers," offered Don. "We can bring 'em down to a place where the velocity-deceleration vectors are perpendicular to the floor upon which we stand while our ship is sluing. We've got a lot of time on our hands, and I, for one, feel a lot happier when I'm doing something."

"It's a thought," said Hadley. "Wanna try it?"

"Let's go."

Thirty hours after the *Relay Girl* left the Station, Walt and Franklen held a council of war, in which Charley Bren was the prime factor.

"We've come about two hundred million miles, and our present velocity is something like four thousand miles per second," said Walt. "We're going out towards Mars on a slightly-off radial course, to the North of the ecliptic. That means we're a little over a quarter of a billion miles from Sol, or about to hit the Asteroid Belt. Thinking it over a little, I think we should continue our acceleration for another thirty hours. What say?"

"The field has shown no change in intensity that I can detect," said Bren. "If they haven't dropped their radiated intensity, that means that we are no closer to them than we were before. Of course, we'd probably have to cut the distance by at least a half before any measurable decrement made itself evident."

"They must be on the upper limit of that four thousand MPS," observed Walt. "There's one thing certain, we'll never catch them by matching their speed."

"Where will another thirty hours at 6-G put us and how fast?" asked Franklen.

Silence ensued while they scribbled long figures on scratch paper.

"About eight hundred million miles from Sol," announced Walt.

"And about eight thousand MPS," added Charley.

"That's a little extreme, don't you think?" asked Franklen.

"By about thirty percent," said Walt, scratching his chin. "If we hold to our original idea of hitting it for six thousand, where will we be?"

"That would make it about forty-five hours from take-off, and we'd be about four hundred and sixty million miles from Sol." Charley grinned widely and said: "By Jove!"

"What?"

"By Jove!"

"By Jove! What?"

"That's where we'd be—By Jove!"

"*Phew.*"

"I agree with you," said Franklen to Walt. "Better ignore him."

"Sure will after that. So then we'll be 'By Jove' at six thousand. That would be a swell place to make turnover, I think. At 1-G decel, to about four thousand MPS, that'll put us about ... um, that'd take us ninety hours! We'll make that 3-G, at twenty hours, which will put us about three hundred and fifty million miles along, which plus the original four hundred and sixty million adds up to eight hundred and ten million miles—"

"When an astronaut begins to talk like that," interrupted Arden, "we of the skyways say that he is talking in Congressional figures. The shoe is on the other foot. What on earth are you fellows figuring?"

"Where we'll be and how fast we'll be going at a given instant of no particular importance," offered Walt. "When did you wake up?"

"About the third hundred million. All of those ciphers going by made a hollow sound, like a bullet whistling in the wind."

"Well, we're trying to make the theories of probability match with figures. We'll know in about forty-five hours whether we were right or not."

"It's a good thing we have all space to go around in. Are you sure that we have all eternity?"

"Don't get anxious. They're still coming in like a ton of bricks four times per hour, which means that they're riding easy. I don't want to overrun them at about three thousand MPS and have to spend a week decelerating, returning, more decelerating, and then matching velocities."

"I see. You know best. And where is this Asteroid Belt that I've heard so much about?"

"To the South of us by a few million miles. Those bright specks that you can't tell from stars are asteroids. The common conception of the Asteroid Belt being filled to overflowing with a collection of cosmic rubble like the rings of Saturn is a lot of hooey. We'll be past in a little while and we haven't even come close to one. Space is large enough for all of us, I think."

"But not when all of us want the same space."

"I don't care for their area," said Walt with a smile. "Let 'em have it, I don't care. I'll stay up here and let them run as they will."

"You mean the ones that are moving downward?" asked Arden, indicating the sky.

"Those are asteroids, yes. We're to the North, as you may check by going around the ship to the opposite side. You'll see Polaris almost directly opposite, there. Sol is almost directly below us, and that bright one that you can see if you squint almost straight up out of the port is Saturn."

"I won't bother crossing the ship to see Polaris. I prefer the Southern Cross anyway. The thing I'm most interested in is: Are we accomplishing anything?"

"I think that we've spent the last thirty hours just catching up," explained Walt. "Up to right now we were going backwards, so to speak: we're on even terms now, and will be doing better from here on in."

"It's the waiting that gets me down," said Arden. "Oh, for something to do!"

"Let's eat," suggested Walt. "I'm hungry, and now that I think of it, I have not eaten since we left the Station. Arden, you are hereby elected to the post of galley chief. Get Jimmy from the dome if you need help."

"Help? What for?"

"He can help you lift it out of the oven. Don must have a cast-iron stomach."

"That's hearsay. I'll show you! As soon as I find the can opener, breakfast will be served."

"Make mine dinner," said Charley. "We've been awake all the time."

"O.K., we will have a combined meal, from grapefruit to ice cream. Those who want any or all parts may choose at will. And fellows, please let me know as soon as you get something tangible."

"That's a promise," said Walt. "Take it easy, and don't worry. We'll be catching up with them one of these days."

"Hadley, how much coating have we got on those cathodes?"

"Not too much. We had about twenty G-hours to begin with. We went to a half-G for about twenty hours, and now we're running on a quarter-G, which would leave us go for forty hours more. That's a grand total of about sixty hours."

"And the batteries?"

"In pretty good shape."

"Well, look. If it should come to a choice between floor and signal gun, we'll choose the gun. We've about twelve hours left in the cathodes, and since everybody is now used to quarter-G we might even slide it down to an eighth-G, which would give us about twenty-four hours."

"Your gun is still putting out?"

"So far as I can tell. Ten hours from now, we should know, I think, predicating my guess on whatever meager information they must have."

"We could save some juice by killing most of the lights in the ship."

"That's a thought. Johannson, have one of your men run around and remove all lights that aren't absolutely necessary. He can kill about three quarters of

them, I'm certain. That'll save us a few kilowatt hours," said Channing. "And another thing. I'm about to drop the power of our electron gun and run it continuously. If the boys are anywhere in the neighborhood, they'll be needing the continuous disturbance for direction finding. I'd say in another five hours that we should start continuous radiation."

"You know, Channing, if this thing works out all right, it will be a definite vote for pure, deductive reasoning."

"I know. But the pure deduction is not too pure. It isn't guesswork. There are two factors of known quantity. One is that I know Walt Franks, and the other is that he knows me. The rest is a simple matter of the boys on the Station knowing space to the last inch, and applying the theory of probabilities to it. We'll hear from them soon, or I'll miss my guess. You wait."

"Yeah," drawled Captain Johansson, "we'll wait!"

Charley Bren made another computation and said: "Well, Walt, we've been narrowing them down for quite a time now. We're getting closer and closer to them, according to the field intensity. I've just got a good idea of direction on that last five-minute shot. Have Franklen swivel us around on this course; pretty soon we'll be right in the middle of their shots."

"We're approaching them asymptotically," observed Walt. "I wish I knew what our velocity was with respect to theirs. Something tells me that it would be much simpler if I knew."

"Walt," asked Arden, "how close can you see a spaceship?"

"You mean how far? Well, I don't know that it's ever been tried and recorded. But we can figure it out easy enough, by analogy. A period is about thirty thousandths of an inch in diameter, and visible from a distance of thirty inches. I mean visible with no doubt about it's being there. That's a thousand to one. Now, the *Solar Queen* is about six hundred feet tall and about four hundred feet in its major diameter, so we can assume a little more than the four hundred feet—say five hundred feet average of circular area, say—follow me?"

"Go on, you're vague, but normal."

"Then at a thousand to one, that becomes five hundred thousand feet, and dividing by five thousand—round figures because it isn't important enough to use that two hundred and eighty feet over the five thousand—gives us one thousand miles. We should be able to see the *Solar Queen* from a distance of a thousand miles."

"Then at four thousand miles per second we'll be in and out of visual range in a half second?"

"Oh no. They're rambling on a quite similar course at an unknown but high velocity. Our velocity with respect to theirs is what will determine how long they're within visual range."

"Hey, Walt," came the voice of Charley Bren. "The intensity of Don's beam has been cut to about one quarter and is now continuous. Does that mean anything?"

"Might mean trouble for them. Either they're running out of soup and mean for us to hurry up, or they assume we're close enough to obviate the need for high power. We'd better assume they want haste and act accordingly. How're the boys on the radio detectors coming along?"

"Fine. They've taken over the direction finding and claim that we are right on their tail."

"Anything in the sights, Jimmy?"

"Not yet. But the electroscope boys claim that quarter power or not, the input is terrific."

"Take a rest, Jimmy. We won't be there for a while yet. No use burning your eyes out trying to see 'em. There'll be time enough for you to do your share after we get 'em close enough to see with the naked eye. What do the beam-scanners say?"

"Shucks," answered the man on the scanners, "they're still radiating. How are we going to fix 'em on a reflected wave when they're more powerful on their own hook? The whole plate is glaring white. And, incidentally, so is the celestial globe in the meteor-spotter. I've had the threshold cut to the devil on that or we'd never be able to hold this course. Anything like a

meteor that comes in our way now will not register until we're right on top of it and—"

The *Relay Girl* lurched sickeningly. All over the ship, things rattled and fell to the floors. Men grabbed at the closest solid object, and then the *Relay Girl* straightened out once more.

"Woosh," said Franks. "That was a big one."

"Big one?" called Charley Bren. "That, my friend, was none other than the *Solar Queen*!"

"Can you prove that?"

"Sure. Our electroscopes now indicate a positive charge; they crossed over just as we lurched."

"Jimmy, get your scope a-top and get looking. Franklen, hang on about 7-G and follow Jimmy's orders. Charley, see if you can get anything cogent out of your gadget. Holy Green Fire, with all of a cubic million million million megaparsecs in which to run, we have to be so good that we run right into our quarry. Who says that radio direction finding is not a precise science? Who says that we couldn't catch—"

"Walt, they're in sight, but losing fast."

"O.K., Jimmy, can you give me any idea as to their velocity with respect to ours?"

"How long is she?"

"Six hundred feet."

Jimmy was silent for some seconds. "They're out of sight again, but I make it about four to seven hundred miles per second."

"At 7-G we should match that seven hundred in about four hours."

"And then go on decelerating so that they'll catch up?"

"No," said Walt. "I used the max figure and we can assume that they aren't going that fast, quite. At the end of four hours, we'll turnover and wait until

they heave in sight again and then we'll do some more oscillating. We can match their velocity inside of ten hours, or Franklen will get fired."

"If I don't," promised Franklen, "I'll quit. You can't fire me!"

"We should be able to contact them by radio," said Walt.

"Their beam is off," said Bren.

"And they are using the landing set," called the radio man, "It's Channing. He says: 'Fancy meeting you here.' Any answer?"

"Just say, 'Dr. Channing, I presume?'"

Channing's voice came out of the ship's announcer system as the radio man made the necessary connections. It said: "Right—but what kept you so long?"

"Our boss was away," said Walt. "And we can't do a thing without him."

"Some boss. Some crew of wild men. Can't go off on a fishing trip without having my bunch chasing all over the Solar System."

"What's wrong with a little sightseeing tour? We didn't mean any harm. And speaking of harm, how are you and the rest of that bunch getting along?"

"We're O.K. What do you plan after we finally get close enough together to throw stones across?"

"We've got a whole hold full of spare batteries and a double set of replacement cathodes. There is a shipload of gravanol aboard, too. You'll need that and so will we. By the time we finish this jaunt, we'll have been about as far out as anybody ever gets."

"Yeah—got any precise figures? We've been running on a guess and a hope. I make it out about seven hundred million."

"Make it eight and a half. At 6-G, you'll cover another hundred and fifty million miles before you stop. Take it twenty-two hours at 6-G—and then another twenty-two at 6. That should put you right back here but going the other way at the same velocity. But wait, you've been coasting. Mark off that last twenty-two hours and make it like this: You'll be one thousand million miles from Sol when you come to a stop at the end of the first twenty-two hours at 6-G. That hangs you out beyond the orbit of Saturn by

a couple of hundred million. Make it back forty-four hours at 6-G, turnover and continue. By that time we'll all be in so close that we can make any planet at will—preferably you to Terra and we'll head for Venus Equilateral. You'll come aboard us? No need for you to go with the rest."

"I can have the scooter sent out," said Channing. "How's Arden?"

"I'm fine, you big runabout. Wait until I get you!"

"Why Arden, I thought you might be glad to see me."

"Glad to see you?"

"But Arden—"

"Don't you 'But Arden' me, you big gadabout. Glad to see you? Boy, any man that makes me chase him all over the Solar System! You just wait. As soon as I get ahold of you, Don Channing, I'm going to—bust out and bawl like a kid! Hurry up, willya?"

"I'll be right over," said Don soberly.

And, strangely enough, Don did not deviate this time.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK OFF THE BEAM

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